Net Force and Acceleration (template)

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

What decides how quickly something speeds up?

Targets

- I can say what net force means.
- I can predict how force and mass change acceleration.
- I can calculate acceleration with a = F / m.

Concepts

- **Net force** (F_net = F1 + F2): Add every force on the object. A push the other way counts as negative.
- **Acceleration** (a = F_net / m): How fast the speed changes. Bigger net force: bigger acceleration. Bigger mass: smaller acceleration.

What the class predicted, and what is true

- A cart is pushed with a steady force. Now the force doubles. What happens to its acceleration? **It doubles.** a = F / m: double F, double a (mass is unchanged).
- The same force pushes a cart. Now the cart carries twice the mass. What happens to its acceleration? **It halves.** a = F / m: double m, half a (force is unchanged).

Huddle notes (answers filled in)

Definitions

- **Net force:** The **sum** of all the forces on an object.
- **Acceleration:** How fast the **velocity** changes.

Worked example

A 2 kg cart is pushed with a net force of 6 N. Find its acceleration.

- a = F / m
- a = 6 / 2 = **3** m/s^2
- In 1 s its speed rises by **3** m/s.

Label the cart

1. The **applied force** pushes to the right.
2. **Friction** pushes to the left.

One worked sample of each skill

Find the acceleration

Level 1. A 3 kg cart is pushed with a net force of 9 N. What is its acceleration?
Answer: 3 m/s² a = F / m = 9 / 3 = 3 m/s².

Level 2. A 0.5 kg cart feels a net force of 3 N. What is its acceleration?
Answer: 6 m/s² a = 3 / 0.5 = 6 m/s².

Level 3. A 5 kg cart is pushed right with 15 N while 5 N pushes left. What is its acceleration?
Answer: 2 m/s² Net force = 15 – 5 = 10 N. a = 10 / 5 = 2 m/s².

Level 4. A rope pulls a 6 kg crate with 21 N. Friction is 3 N. What is the acceleration?
Answer: 3 m/s² Net = 21 – 3 = 18 N. a = 18 / 6 = 3 m/s².

Find the force

Level 1. A 3 kg cart accelerates at 3 m/s². What net force acts on it?
Answer: 9 N F = m a = 3 × 3 = 9 N.

Level 2. A 0.5 kg cart accelerates at 4 m/s². What net force acts on it?
Answer: 2 N F = 0.5 × 4 = 2 N.

Level 3. A 2500 g cart accelerates at 2 m/s². What net force acts on it, in newtons?
Answer: 5 N 2500 g = 2.5 kg. F = 2.5 × 2 = 5 N.

Level 4. A 4 kg cart starts from rest and reaches 6 m/s in 2 s. What net force acted on it?
Answer: 12 N a = 6 / 2 = 3 m/s². F = 4 × 3 = 12 N.